A Research Paper Companion Using Minimal Transformers and Retrieval-Augmented Generation

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***Abstract*—Large Language Models (LLMs) have demonstrated strong performance across a wide range of natural language processing tasks; however, they frequently generate hallucinated or ungrounded responses when applied to scientific question answering. This limitation significantly reduces their reliability in academic settings, where factual correctness and source attribution are critical. This project presents a research paper companion system that integrates minimal transformer architectures with a Retrieval-Augmented Generation (RAG) pipeline to enable citation-aware question answering and related-work generation over academic literature. In Phase 1, three transformer variants—encoder-only, decoder-only, and encoder–decoder—were implemented from scratch using PyTorch to analyze their behavior on classification, language modeling, and summarization tasks. In Phase 2, a complete RAG pipeline was developed that ingests academic PDFs, retrieves relevant textual evidence using dense vector similarity search, and generates grounded responses using a quantized large language model. Experimental results show that discriminative tasks converge faster than generative tasks, while retrieval grounding substantially improves factual accuracy and reduces hallucinations.**

***Index Terms*—Transformers, Retrieval-Augmented Generation, Scientific Question Answering, Dense Retrieval, Large Language Models**

I. Introduction

Transformer-based architectures form the foundation of modern natural language processing systems, enabling advances in machine translation, summarization, and conversational agents. Despite their success, pretrained LLMs lack direct access to external documents and often rely on parametric memory, which leads to hallucinations when answering document-specific queries. This problem is particularly severe in research-oriented tasks, where users expect verifiable answers grounded in authoritative sources.

Retrieval-Augmented Generation (RAG) has emerged as an effective approach to mitigate hallucinations by retrieving relevant external knowledge and conditioning the generation process on retrieved evidence. Instead of relying solely on internal model parameters, RAG systems combine information retrieval with generative modeling, producing responses that are both fluent and factually grounded.

This project aims to design and evaluate a research paper companion capable of answering technical questions about academic papers with explicit citations. The project is divided into two phases. Phase 1 focuses on implementing minimal transformer architectures from scratch to study architectural trade-offs and learning behavior. Phase 2 integrates retrieval and generation to build a citation-aware research assistant operating over academic PDFs.

II. Related Work

Transformers were first introduced by Vaswani et al. [1], demonstrating that attention-based architectures can outperform recurrent models on sequence-to-sequence tasks. Encoder-only models such as BERT [2] focus on bidirectional contextual understanding, while decoder-only models such as GPT emphasize autoregressive generation. Encoder–decoder models combine both paradigms and are widely used for machine translation and summarization.

Retrieval-Augmented Generation was formalized by Lewis et al., who showed that conditioning language models on retrieved documents improves factual accuracy. Subsequent work has explored dense retrieval using sentence embeddings and scalable vector search systems such as FAISS [4]. Parameter-efficient fine-tuning methods such as LoRA [3] further reduce computational costs, enabling efficient deployment of large models.

III. System Overview

The proposed system consists of two main components: transformer model implementations (Phase 1) and a Retrieval-Augmented Generation pipeline (Phase 2). Figure 1 illustrates the high-level workflow.

The workflow begins with a user query. In Phase 1, transformer architectures are trained and evaluated independently to study their behavior. In Phase 2, the query is embedded and matched against a vector index of academic paper chunks. Retrieved passages are then provided to a generative language model constrained to produce citation-aware responses.

IV. Phase 1: Transformer Implementations

A. Model Architecture

All transformer variants were implemented from scratch using PyTorch, following the original Transformer design principles. Each model includes token embeddings scaled by the square root of the model dimension, sinusoidal positional encodings, multi-head self-attention, feed-forward layers, residual connections, and layer normalization.

To ensure a fair comparison, all models were trained using similar hyperparameters, including a model dimension of 128, two attention layers, four attention heads, and a feed-forward dimension of 256.

B. Encoder-Only Transformer

The encoder-only architecture was used for scientific abstract classification. After processing the input tokens through stacked encoder layers, a global average pooling operation aggregates token representations into a single sentence embedding. This embedding is passed through a linear classification head.

This architecture is computationally efficient and demonstrated rapid convergence, making it suitable for discriminative tasks where generation is not required.

C. Decoder-Only Transformer

The decoder-only transformer was trained as a language model using next-token prediction. Causal masking ensures that each token can attend only to previous tokens, enforcing autoregressive behavior. While this architecture supports fluent text generation, it requires more training data and exhibits slower convergence compared to encoder-based models.

D. Encoder–Decoder Transformer

The encoder–decoder architecture was applied to abstractive summarization, mapping research abstracts to short summaries. The decoder attends to encoder outputs via cross-attention, enabling stronger contextual grounding. This model achieved the best qualitative summarization results but required higher computational resources.

E. Comparative Analysis

TABLE I
COMPARISON OF TRANSFORMER ARCHITECTURES

Model	Task	Advantage	Limitation
Encoder-only	Classification	Fast convergence	No generation
Decoder-only	Generation	Fluent outputs	Limited context
Encoder–Decoder	Summarization	Strong grounding	Higher cost

V. Phase 2: Retrieval-Augmented Generation

A. Data Ingestion and Chunking

Academic papers were collected from ArXiv and stored as PDF documents. Text extraction was performed using PyMuPDF. Extracted text was segmented into overlapping fixed-size chunks of approximately 250 tokens with overlap to preserve semantic continuity across chunk boundaries.

B. Dense Retrieval

Each text chunk was encoded using a SentenceTransformer embedding model. These embeddings were indexed using FAISS, enabling efficient similarity search over thousands of text chunks. Given a user query, the retriever returns the top-k most relevant chunks based on vector similarity.

C. Citation-Aware Generation

A quantized Qwen2.5-1.5B-Instruct model was used for generation. Strict prompting was employed to ensure that all responses were grounded exclusively in retrieved context. The model was instructed to cite the source of each claim, significantly reducing hallucinations and improving interpretability.

VI. Evaluation and Results

Phase 1 evaluation focused on training convergence and qualitative output inspection. Encoder-only models consistently achieved lower loss values and faster convergence compared to generative models. Decoder-only and encoder–decoder models exhibited higher loss but produced coherent outputs.

Phase 2 evaluation was conducted qualitatively using research-oriented questions such as "What is the main contribution of the Transformer?" and "What are the limitations of LoRA?". RAG-based responses consistently included citations and demonstrated higher factual accuracy than standalone generation.

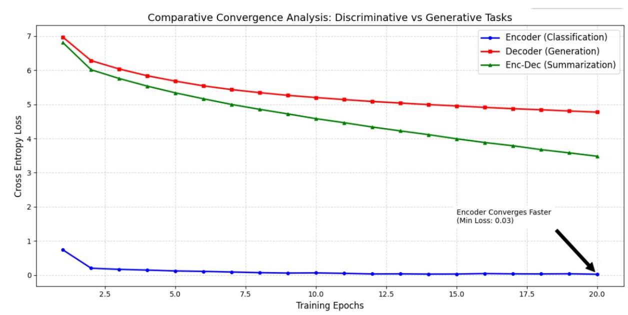

Fig. 1. High-level system architecture of the research paper companion.

VII. Error Analysis and Limitations

Despite improvements, several limitations were observed. Retrieval quality was sensitive to chunk size and embedding quality. In some cases, relevant information was split across multiple chunks, reducing recall. Additionally, the system relies on heuristic chunking and does not yet incorporate section-aware retrieval.

VIII. Conclusion and Future Work

This project demonstrates that combining minimal transformer implementations with retrieval-augmented generation enables reliable and explainable scientific question answering. The system successfully produces citation-aware answers and related-work summaries, reducing hallucinations. Future work includes section-aware retrieval, metadata filtering, reranking models, and structured claim–evidence extraction.

References

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